

MULTI-STEP AI WORKFLOW

Assessment Submission

Scenario: AI Study Planner — From Student Requirements to a Refined Personalized Plan

Student Name	THUMMAPUDI MANIDEEP
Registration Number	99230040438
Email	99230040438@klu.ac.in
Department	CSE-AIML
Assessment Format	Chained multi-prompt AI workflow

Workflow at a Glance

Requirements · Planning Blueprint · Weekly Timetable · Critique & Refinement. The workflow aims to produce a realistic study plan that balances technical subjects, project work, breaks and revision.

1. Assessment Understanding & Workflow Design

The task is to design a multi-stage AI workflow using at least three connected prompts. Each prompt builds on the prior output, and one stage critiques and refines the result before the final answer is accepted.

Selected Scenario

AI Study Planner: Convert a student's academic constraints into a practical weekly study plan, generate a detailed schedule from that plan, then audit and improve the schedule against explicit criteria.

Stage	Purpose	Input	Output
1 — Planning	Extract priorities, constraints and planning rules.	Student requirements	Structured study-plan blueprint
2 — Generation	Turn the blueprint into a usable weekly timetable.	Stage 1 blueprint	Detailed weekly schedule
3 — Critique & Refinement	Audit the schedule and correct weaknesses.	Stage 2 schedule + checklist	Critique + refined schedule

Why this scenario is suitable

A useful timetable should not be generated blindly. Planning identifies priorities, generation turns those choices into a schedule, and critique checks whether the result is balanced, realistic and complete.

Design Principles

- Traceability: each stage consumes a previous output.
- No premature solution: requirements are analyzed first.
- Testability: the schedule can be checked against constraints.
- Refinement: weaknesses are corrected before finalization.

Practical Planning Considerations

A strong plan should account for class hours, assignment deadlines, energy levels and available study resources. If a task takes longer than expected, move it to the buffer rather than removing daily

2. Exact Prompt Chain Transcript

The prompts are designed to run in the same AI conversation. Each later prompt explicitly uses the previous stage's output.

Prompt 1 — Extraction & Planning

Act as an expert academic planning assistant. I am a CSE-AI/ML student preparing for multiple subjects and technical interviews. I have 7 days available. My priorities are Java programming, SQL/DBMS, DSA, AI/ML fundamentals, and project work. I can study about 4 focused hours on weekdays and 6 hours on the weekend. I prefer shorter sessions with breaks and daily revision.

Task: Extract requirements, identify priorities, create a structured weekly planning blueprint, and assign a reasonable time allocation to each area. Do not create the final timetable yet. Present a table followed by 4-6 planning rules.

Prompt 2 — Generation / Drafting

Using only the planning blueprint and rules from Step 1, create the first draft of a 7-day study timetable. Preserve priorities and time allocations. Constraints: 4 focused hours Monday-Friday and 6 hours Saturday-Sunday; include breaks, daily revision, all listed subjects, practical session lengths, and a small project buffer. Present a day-by-day table and a short strategy explanation.

Prompt 3 — Critique Gate & Refinement

Act as a strict academic reviewer. Review the Step 2 draft for: (1) weekday/weekend hour limits, (2) meaningful time for all five priority areas, (3) daily revision, (4) reasonable session lengths, (5) project buffer, and (6) a logical learning sequence. First list weaknesses and corrections. Then output the complete corrected timetable, improving Step 2 rather than replacing it without reason.

Expected Output from the Prompt Chain

Stage 1 makes the requirements explicit; Stage 2 converts them into time blocks; Stage 3 checks constraints and corrects weaknesses. Keeping each output visible makes the process easier to evaluate and

3. Intermediate AI Responses

Step 1 Output — Planning Blueprint

Area	Priority	Weekly Focus	Planning Rule
Java	High	7–8 hrs	Practice code daily; increase difficulty gradually.
DSA	High	6–7 hrs	Combine concept review with problem solving.
SQL / DBMS	High	5–6 hrs	Alternate SQL practice and DBMS concepts.
AI / ML	Medium	4–5 hrs	Use concept → example → short recall pattern.
Project	Medium	3–4 hrs	Keep a protected buffer instead of one large block.

Planning rules: (1) Do difficult work earlier. (2) Use 50–60 minute focus blocks with short breaks. (3) End each day with active recall. (4) Avoid long runs of similar topics. (5) Keep project work manageable. (6) Use weekends for deeper practice and consolidation.

Step 2 Output — First Draft Schedule

Day	Focus Blocks	Revision / Buffer
Mon	Java practice • DSA concepts • SQL practice • AI/ML basics	20 min Java + SQL recall
Tue	DSA problems • Java OOP • DBMS concepts • Project	20 min DSA recall
Wed	SQL queries • Java coding • AI/ML concepts • DSA problems	20 min mixed recall
Thu	DSA problems • DBMS • Java coding • Project	20 min Java/DBMS recall
Fri	Java interview practice • SQL • AI/ML • DSA	20 min weak-area recall
Sat	Deep DSA • Java project coding • SQL/DBMS • AI/ML	30 min revision + project buffer
Sun	Mock coding • SQL test • AI/ML review • Project • Weekly recap	Weekly review + next-week notes

How to Use the Blueprint

Treat weekly-hour estimates as guidance, not rigid quotas. At the end of each study block, record what was completed and what remains. Use that short record to adjust the next session without disrupting the

4. Critique Gate & Final Refined Output

Step 3 Self-Critique

Checklist	Review	Action
Hour limits	Revision may exceed the focused-hour target if not counted.	Count revision within daily focus time.
Priority areas	Project work needs consistent time.	Protect project blocks on weekdays and weekend.
Daily revision	Revision is present each day.	Use active recall instead of rereading.
Session length	Draft lacks exact block lengths.	Use 50-minute focus + 10-minute break cycles.
Learning sequence	Concepts and practice can connect better.	Place concepts before related practice.

Final Corrected 7-Day Study Plan

Day	Focus Sessions	Breaks / Review
Monday 4 hrs	Java coding; DSA concepts + 2 problems; SQL query practice; AI/ML fundamentals	Breaks between sessions; 20 min active recall
Tuesday 4 hrs	DSA problems; Java OOP; DBMS concepts; project implementation	Short breaks; DSA + DBMS recall
Wednesday 4 hrs	SQL practice; Java coding; AI/ML concepts + examples; DSA problems	Short breaks; mixed recall
Thursday 4 hrs	DSA problems; DBMS + SQL; Java interview questions; project implementation	Short breaks; Java/DBMS recall
Friday 4 hrs	Java mock coding; SQL test; AI/ML revision; DSA timed practice	Short breaks; identify weak areas
Saturday 6 hrs	Deep DSA practice; Java problem set; SQL/DBMS test; AI/ML application; project work; weak-area repair	Break after each block; 30 min consolidation

Monitoring Progress

Track completed sessions, practice questions solved, concepts recalled without notes and project milestones reached. These indicators provide more useful feedback than study time alone.

4. Critique Gate & Final Refined Output (continued)

Day	Focus Sessions	Breaks / Review
Sunday 6 hrs	Coding mock test; SQL/DBMS mock test; AI/ML review; project completion; DSA revision; weekly recap + next-week planning	Breaks after blocks; 30 min recall + planning

Refinement Result

The final schedule retains the original priorities, converts them into concrete sessions, adds protected project time, uses consistent focus cycles, and makes daily revision explicit. The critique stage improves the schedule before it is accepted.

Decision Rule

At the end of the week, check which sessions were completed, whether each subject received time, and whether daily recall identified weak areas. Adjust the next week's allocation using observed progress.

Key takeaway

A chained AI workflow separates planning, generation and critique. The final answer is reviewed against clear criteria instead of being accepted after a single prompt.

Weekly Review Method

At the end of Sunday, identify two strong areas, two weak areas and one priority for the next week. Carry unfinished tasks forward selectively so that the new plan remains achievable.

5. Documentation, Dependency Check & Submission Evidence

This section explains how the assessment meets the chained-workflow requirements and identifies evidence that should accompany the submission if required.

Dependency Verification

Requirement	How this submission satisfies it
At least 3 chained prompts	Three prompts are included: planning → generation → critique/refinement.
Explicit dependency	Prompt 2 uses the Step 1 blueprint; Prompt 3 audits and improves Step 2.
Critique gate	Prompt 3 uses six criteria before producing the corrected timetable.
Intermediate outputs	The planning blueprint and first draft are documented before the final result.
Live execution	Run the prompts sequentially in the assigned AI interface.
Submission evidence	Attach a genuine screenshot or chat link if direct interface evidence is required.

Live Evidence

Insert live AI-chat screenshot here

Recommended evidence: a screenshot showing Prompt 1 and its response, Prompt 2 and its response, Prompt 3, and the final refined response. Use a genuine screenshot from the assigned AI interface.

Final checklist: verify student details → verify all three prompts → verify intermediate responses → add live screenshot/chat link if required → submit as PDF.

End of Assessment Submission

Final Quality Check

Before submission, confirm that the three prompts are connected, intermediate outputs are documented, the critique leads to visible changes, and the student details on the cover are correct. Add genuine live-